

Quadratic Equations

Chapter 4 - Hinglish Notes, Methods, Examples + Solutions

Ye Chapter 4 (Quadratic Equations) ka complete guide hai. Pehle red-page ke core basics (power, coefficient, zero product rule, +/- etc.), fir chapter ka main content - factorisation, quadratic formula, discriminant aur sum/product of roots. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Power / Square

- $x^2 = x$ times x . Ise 'x square' bolte hain.
- x^2 me x base hai aur 2 power hai.

2. Equation vs Expression

EXAM * Exam me CONSTANTLY aata hai

Expression Sirf likha hua, '=' nahi. Jaise $2x^2 + 3x - 5$.

Equation: '=' sign ke saath. Jaise $2x^2 + 3x - 5 = 0$.

3. Degree

- Degree = sabse bada power. Quadratic me degree hamesha 2 hota hai.

4. Coefficient (a, b, c)

EXAM * Exam me CONSTANTLY aata hai

Rule: $ax^2 + bx + c$ me $a = x^2$ ka, $b = x$ ka, $c = \text{constant}$.

- Example: $3x^2 - 5x + 2 \rightarrow a = 3, b = -5, c = 2$.

5. Root / Zero / Solution

EXAM * Exam me CONSTANTLY aata hai

Definition Root = wo x jiske liye equation 0 ho jaye.

- Quadratic equation ke MAXIMUM 2 roots hote hain.

6. Factorisation (Splitting Middle Term)

EXAM * Exam me CONSTANTLY aata hai

Idea: $ax^2 + bx + c$ ko $(..)(..)$ me todo.

- Middle term b ko aise 2 numbers p, q me todo jahan:
- $p + q = b$ AND $p \times q = a \times c$.

7. Square Root (sqrt)

- $\text{sqrt}(25) = 5$, kyunki $5 \times 5 = 25$. Ye power (square) ka ulta hai.
- $\text{sqrt}(D)$ ka matlab D ka square root.

8. Zero Product Rule

EXAM * Exam me CONSTANTLY aata hai

Rule: Agar $A \times B = 0$, to ya $A = 0$ ya $B = 0$ (ya dono).

- Yahi factorisation method ka base hai. $(x+2)(x+3)=0 \rightarrow x=-2$ ya $x=-3$.

9. Plus-Minus (+/-)

EXAM * Exam me CONSTANTLY aata hai

- Quadratic formula me +/- ka matlab DO answers milte hain:
- ek '+' wala root, aur ek '-' wala root.

PART B - Chapter 4 ka Main Content

10. Quadratic Equation Kya Hai?

EXAM * Exam me CONSTANTLY aata hai

Standard Form $ax^2 + bx + c = 0$, jahan $a \neq 0$.

- $a \neq 0$ zaroori hai (warna x^2 hi nahi bachega, linear ban jayega).
- Example: $2x^2 - 3x + 1 = 0$.

11. Solve Karne ke Methods

EXAM * Exam me CONSTANTLY aata hai

(a) Factorisation Middle term todo, fir Zero Product Rule lagao.

(b) Quadratic Formula Hamesha kaam karta hai:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Yahan $b^2 - 4ac$ ko Discriminant (D) kehte hain.

(c) Completing the Square Equation ko $(x + k)^2 = \text{number}$ form me badal kar solve karna.

12. Discriminant (D) aur Nature of Roots

EXAM * Exam me CONSTANTLY aata hai

Formula: $D = b^2 - 4ac$.

$D > 0 \rightarrow$ 2 alag-alag (distinct) real roots

$D = 0 \rightarrow$ 2 barabar (equal) real roots

$D < 0 \rightarrow$ koi real root nahi (imaginary)

13. Sum aur Product of Roots

EXAM * Exam me CONSTANTLY aata hai

Agar roots A aur B hon:

- Roots ka SUM ($A + B$) = $-b/a$
- Roots ka PRODUCT ($A \times B$) = c/a
- Example: $2x^2 - 7x + 3 = 0 \rightarrow \text{sum} = 7/2$, product = $3/2$.

14. Word Problems

- Sentence ko x me likho, equation banao, fir solve karo.
- Common: consecutive numbers, area/length, age, speed-time problems.
- Answer me practical value lo (jaise length negative nahi hoti).

Quick Revision

Standard form: $ax^2 + bx + c = 0$ ($a \neq 0$).

Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

$D = b^2 - 4ac$: >0 distinct, $=0$ equal, <0 no real root.

Sum of roots = $-b/a$, Product = c/a .

Factorise: $p+q=b$, $p \times q = a \times c$, fir zero product rule.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1 (Factorisation): $x^2 + 5x + 6 = 0$

- Step: $a=1$, $b=5$, $c=6$. $a \times c = 6$.
- Step: $p+q=5$ aur $pxq=6 \rightarrow 2, 3$.
- Step: $x^2 + 2x + 3x + 6 = 0 \rightarrow (x+2)(x+3) = 0$.
- Step: $x + 2 = 0$ ya $x + 3 = 0$.

Answer: $x = -2$ ya $x = -3$

Example 2 (Factorisation): $x^2 - 7x + 12 = 0$

- Step: $p+q = -7$ aur $pxq = 12 \rightarrow -3, -4$.
- Step: $(x - 3)(x - 4) = 0$.

Answer: $x = 3$ ya $x = 4$

Example 3 (Quadratic Formula): $x^2 - 4x + 1 = 0$

- Step: $a=1$, $b=-4$, $c=1$. $D = b^2 - 4ac = 16 - 4 = 12$.
- Step: $x = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm 2\sqrt{3}}{2}$.

Answer: $x = 2 \pm \sqrt{3}$

Example 4 (Discriminant): $x^2 + 2x + 5 = 0$ ke roots ka nature?

- Step: $D = b^2 - 4ac = 4 - 20 = -16$.
- Step: $D < 0$.

Answer: Koi real root nahi

Example 5 (Discriminant): $x^2 - 6x + 9 = 0$ ka nature?

- Step: $D = 36 - 36 = 0$.
- Step: $D = 0 \rightarrow 2$ barabar roots.

Answer: 2 equal real roots ($x = 3$)

Example 6 (Zero Product): $(x - 2)(x + 5) = 0$

- Step: Zero product rule: koi ek bracket 0 hoga.
- Step: $x - 2 = 0$ ya $x + 5 = 0$.

Answer: $x = 2$ ya $x = -5$

Example 7 (Sum & Product): $2x^2 - 7x + 3 = 0$

- Step: $a=2, b=-7, c=3$.
- Step: Sum = $-b/a = 7/2$; Product = $c/a = 3/2$.

Answer: Sum = $7/2$, Product = $3/2$

Example 8 (Square root): $x^2 = 49$

- Step: Dono taraf square root lo.
- Step: $x = \pm \sqrt{49}$.

Answer: $x = 7$ ya $x = -7$

Example 9 (a, b, c pehchaano): $3x^2 - 5x + 2 = 0$

- Step: x^2 ka coefficient a, x ka b, constant c.

Answer: $a = 3, b = -5, c = 2$

Example 10 (Word problem): Do consecutive positive integers ka product 20.

- Step: Maano x aur $x+1$: $x(x+1) = 20 \rightarrow x^2 + x - 20 = 0$.
- Step: $(x + 5)(x - 4) = 0 \rightarrow x = 4$ (positive lete hain).

Answer: Numbers = 4 aur 5

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1. Quadratic equation ka standard form kya hai?
- Q2. Quadratic equation me maximum kitne roots hote hain?
- Q3. $2x^2 + 3x - 5$ expression hai ya equation?
- Q4. $ax^2 + bx + c = 0$ me agar $a = 0$ ho to kya hoga?
- Q5. Discriminant ka formula kya hai?
- Q6. $D > 0$ ho to roots kaise hote hain?
- Q7. $D = 0$ ho to roots kaise hote hain?
- Q8. $D < 0$ ho to roots kaise hote hain?
- Q9. Quadratic formula likho.
- Q10. Solve karo: $x^2 - 9 = 0$.
- Q11. Solve karo: $x^2 + 7x + 10 = 0$.
- Q12. $3x^2 - 5x + 2 = 0$ me a, b, c batao.
- Q13. Zero product rule kya kehta hai?
- Q14. Roots ka sum ($A + B$) kis ke barabar hota hai?
- Q15. Roots ka product ($A \times B$) kis ke barabar hota hai?
- Q16. Solve karo: $(x - 3)(x + 4) = 0$.
- Q17. $x^2 = 36$ ho to $x = ?$
- Q18. $x^2 - 4x + 4 = 0$ ka D aur nature?
- Q19. Splitting middle term me kaunsi 2 conditions chahiye?
- Q20. $x^2 - 5x + 6 = 0$ ke roots kya hain?

Step-by-Step Solutions

Q1. Standard form?

- Step: x^2 wala term, $a \neq 0$.

Answer: $ax^2 + bx + c = 0$

Q2. Max kitne roots?

- Step: Degree 2 $\rightarrow$ maximum 2 roots.

Answer: 2

Q3. $2x^2 + 3x - 5$ expression ya equation?

- Step: '=' sign nahi hai.

Answer: Expression

Q4. $a = 0$ ho to?

- Step: x^2 term hat jata hai $\rightarrow$ bachta hai $bx + c$.

Answer: Linear ban jata hai

Q5. Discriminant formula?

- Step: b^2 me se $4ac$ ghatao.

Answer: $D = b^2 - 4ac$

Q6. $D > 0$?

- Step: 2 alag real roots.

Answer: Distinct real roots

Q7. $D = 0$?

- Step: 2 barabar real roots.

Answer: Equal real roots

Q8. $D < 0$?

- Step: Real roots possible nahi.

Answer: No real roots

Q9. Quadratic formula?

- Step: Standard formula yaad karo.

Answer: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Q10. $x^2 - 9 = 0$?

- Step: $x^2 = 9 \rightarrow x = \pm \sqrt{9}$.

Answer: $x = 3$ ya -3

Q11. $x^2 + 7x + 10 = 0$?

- Step: $p+q=7$, $pxq=10 \rightarrow 2,5$.

- Step: $(x+2)(x+5)=0$.

Answer: $x = -2$ ya -5

Q12. $3x^2 - 5x + 2 = 0$ me a, b, c ?

- Step: x^2 , x aur constant ke coefficients.

Answer: $a=3$, $b=-5$, $c=2$

Q13. Zero product rule?

- Step: $A \times B = 0$ ho to.

Answer: $A = 0$ ya $B = 0$

Q14. Roots ka sum?

- Step: Formula yaad karo.

Answer: $-b/a$

Q15. Roots ka product?

- Step: Formula yaad karo.

Answer:c/a

Q16. $(x - 3)(x + 4) = 0$?

- Step: $x-3=0$ ya $x+4=0$.

Answer: $x = 3$ ya -4

Q17. $x^2 = 36$?

- Step: Dono taraf sqrt: $x = +/- 6$.

Answer: $x = 6$ ya -6

Q18. $x^2 - 4x + 4 = 0$ ka D aur nature?

- Step: $D = 16 - 16 = 0$.

Answer: $D=0$, equal real roots ($x=2$)

Q19. Splitting middle term ki 2 conditions?

- Step: Do numbers p, q chahiye.

Answer: $p+q=b$ AND $p \times q = a \times c$

Q20. $x^2 - 5x + 6 = 0$ ke roots?

- Step: $(x-2)(x-3)=0$.

Answer: $x = 2$ ya 3